Name: Manasi Dinesh Unnithan
SRN: PES2UG24CS260
GitHub Repo: https://github.com/idwmu/PES2UG24CS260-pes-vcs.git

1A

```
manasi@admin:~/PES2UG24CS260-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
manasi@admin:~/PES2UG24CS260-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B

```
manasi@admin:~/PES2UG24CS260-pes-vcs$ find .pes/objects -type f
.pes/objects/17/d49b4d8a23e594eb3b82b3fb72b9203936afd14aadfbdeedc9690fcb8aa752
.pes/objects/fe/76325aa5521b207ebe01e12fd8e9e3abf030cacd5398e3744a3a56a81ad1bd
.pes/objects/21/01b6a05d8f6e113e1ffdf89f155d4524e98b7f20ea2a584bb6827c04533b37
.pes/objects/37/1aac45c3ba401d13c2ce98c9537b4e53c6049b950bf8a33cbc400712d9f4fa
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/70/057cbf1c15983102c20962f28b91640236fd365e601e6f8a943e2253dbc90a
manasi@admin:~/PES2UG24CS260-pes-vcs$
```

2A

```
^[[Amanasi@admin:~/PES2UG24CS260-pes-v./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
manasi@admin:~/PES2UG24CS260-pes-vcs$ git add .
```

2B

```
manasi@admin:~/PES2UG24CS260-pes-vcs$ xxd .pes/objects/17/d49b4d8a23e594eb3b82b3fb72b9203936afd14aadfbdeedc9690fcb8aa752
00000000: 7472 6565 2034 3500 3130 3036 3434 2061  tree 45.100644 a
00000010: 2e74 7874 00fe 7632 5aa5 521b 207e be01  .txt..v2Z.R. ~..
00000020: e12f d8e9 e3ab f030 cacd 5398 e374 4a3a  ./.....0..S..tJ:
00000030: 56a8 1ad1 bd                             V....
manasi@admin:~/PES2UG24CS260-pes-vcs$
```

3A

```
manasi@admin:~/PES2UG24CS260-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
cat .pes/index    # Human-readable text format
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
  staged:     a.txt
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.c
  untracked:  commit.c
  untracked:  test_objects.c
  untracked:  test_tree.c
  untracked:  tree.h
  untracked:  tree.c
  untracked:  README.md
  untracked:  object.c.save
  untracked:  commit.h
  untracked:  .gitignore
  untracked:  pes.c
  untracked:  object.c
  untracked:  pes.h
  untracked:  Makefile
  untracked:  test_sequence.sh
  untracked:  index.h

100644 fe76325aa5521b207ebe01e12fd8e9e3abf030cacd5398e3744a3a56a81ad1bd 1776748573 12 a.txt
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776749862 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776749862 6 file2.txt
manasi@admin:~/PES2UG24CS260-pes-vcs$
```

3B

```
manasi@admin:~/PES2UG24CS260-pes-vcs$ cat .pes/index
100644 fe76325aa5521b207ebe01e12fd8e9e3abf030cacd5398e3744a3a56a81ad1bd 1776748573 12 a.txt
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776749862 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776749862 6 file2.txt
manasi@admin:~/PES2UG24CS260-pes-vcs$
```

4A

```
echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: 667633bd2a00... Initial commit
Committed: efa6bbe13232... Add world
Committed: 582dee676a40... Add farewell
commit 582dee676a40c7a1f9fc542add378a53513989f966293aefc1c374b537171b93
Author: Manasi Dinesh PES2UG24CS260
Date:   1776749927

    Add farewell

commit efa6bbe1323203ba1594c1af93865c1b1dbb80a8c04e9967ac9c909d160106b9
Author: Manasi Dinesh PES2UG24CS260
Date:   1776749927

    Add world

commit 667633bd2a003ef90a5da36382949a7e6d328ba1567688436fd4c88c79ccfd4c
Author: Manasi Dinesh PES2UG24CS260
Date:   1776749927

    Initial commit farewell

commit 70057cbf1c15983102c20962f28b91640236fd365e601e6f8a943e2253dbc90a
Author: Manasi Dinesh PES2UG24CS260
Date:   1776748573

    secondrld

commit 2101b6a05d8f6e113e1ffdf89f155d4524e98b7f20ea2a584bb6827c04533b37
Author: Manasi Dinesh PES2UG24CS260
Date:   1776748573

    first8573

firstCSQ
```

4B

```
manasi@admin:~/PES2UG24CS260-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/17/d49b4d8a23e594eb3b82b3fb72b9203936afd14aadfbdeedc9690fcb8aa752
.pes/objects/21/01b6a05d8f6e113e1ffdf89f155d4524e98b7f20ea2a584bb6827c04533b37
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/37/1aac45c3ba401d13c2ce98c9537b4e53c6049b950bf8a33cbc400712d9f4fa
.pes/objects/58/2dee676a40c7a1f9fc542add378a53513989f966293aefc1c374b537171b93
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/66/7633bd2a003ef90a5da36382949a7e6d328ba1567688436fd4c88c79ccfd4c
.pes/objects/70/057cbf1c15983102c20962f28b91640236fd365e601e6f8a943e2253dbc90a
.pes/objects/a0/fe749a8c4ad8a3081931e906f9ed3d2c60fd3a2d95fe45d31350925d813e65
.pes/objects/a8/a86cc0d4322de67e1f3384e7ceb864a1062fbde567cdbfe7dadbad5ca99cae
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/ef/45d7eeb2b6716eb58170a2f2eaaba77f14c7a6c493e9db1a2740743184ad88
.pes/objects/ef/a6bbe1323203ba1594c1af93865c1b1dbb80a8c04e9967ac9c909d160106b9
.pes/objects/fe/76325aa5521b207ebe01e12fd8e9e3abf030cacd5398e3744a3a56a81ad
.pes/refs/heads/main
```

4C

```
manasi@admin:~/PES2UG24CS260-pes-vcs$ cat .pes/refs/heads/main
582dee676a40c7a1f9fc542add378a53513989f966293aefc1c374b537171b93
manasi@admin:~/PES2UG24CS260-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Q5.1
pes checkout reads the branch file in .pes refs heads to get the target commit hash and updates .pes head to point to that branch reference. it then loads the commit tree and reconstructs the working directory by writing blobs to files and removing files not present in the target tree while also updating the index to match. the complexity comes from handling differences safely because modified or untracked files may be overwritten leading to data loss if not checked

Q5.2
to detect a dirty working directory you compare each tracked file against the index using metadata like mtime size or recomputed hashes. if a file differs from the index and also differs between the current commit tree and the target branch tree then switching would overwrite local changes. in that case checkout must refuse because the working state is inconsistent with the committed state

Q5.3
in detached head head points directly to a commit instead of a branch so new commits are created without updating any branch reference. these commits remain in the object store but become unreachable once head moves away making them easy to lose. they can be recovered by finding their hash through logs or ref-log and creating a new branch pointing to them

Q6.1
Garbage collection starts from all branch heads and traverses commits trees and blobs marking each reachable object recursively. a hash set is used to store visited object ids so traversal is efficient and avoids duplicates. after marking all reachable objects any object not in the set can be safely deleted and the number of objects visited scales roughly with total stored objects which may be several times the number of commits

Q6.2
Running garbage collection during a commit is dangerous because new objects may be written but not yet referenced by a branch. gc may see them as unreachable and delete them before the commit finishes causing corruption. git avoids this by using safe update ordering temporary references and coordination so objects are only collected after they are fully linked into the repository history